ISIC MULTISPECIALITY HOSPITAL

Website Redevelopment

Formal Scope of Work - Additional Functional Modules

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1. Executive Summary

This document sets out the expanded development scope for the ISIC Multispeciality Hospital website, covering every functional module previously discussed and benchmarked against four leading Indian multi-specialty hospital chains consistently share a core set of sections and capabilities, and this scope brings the ISIC site to that standard while emphasising ISIC's particular strengths in spine care, orthopaedics, multi-specialty expansion, international patients, and rehabilitation.

The document is organised in six parts: a benchmark comparison against the reference hospitals, a detailed breakdown of all 25 requested functional modules with technical notes and indicative effort estimates, a dedicated section on the agentic AI capabilities to be built, a recommended three-phase implementation plan, a consolidated effort and timeline summary, and closing assumptions and next steps.

Effort estimates in this document are indicative, at a scoping level of accuracy, and are intended to support prioritisation and budgeting conversations. They should be refined into committed estimates once wireframes, integration contracts (particularly with the HIS and any pharmacy partner), and a delivery team are confirmed.

2. Benchmark Analysis

The table below summarises the sections and capabilities observed across and along with the corresponding recommendation for ISIC.

Priority	Section / Feature	Recommendation for ISIC
Critical	Book Appointment / Find a Doctor + real-time slots	Full module
Critical	Specialties / Centres of Excellence	Spine, Ortho, Cardiac, Renal, Neuro, Gastro, Sports Injury, Rehab, etc.
Critical	Health Check-up Packages (Preventive)	Full module
Critical	International Patients section	Full module
Critical	Patient Stories / Testimonials	Full module
Critical	Health Library / Blogs / Disease & Conditions A-Z	Full module
High	Request a Callback / Second Opinion	Multiple forms
High	Home Care / Homecare Services	Full module

Priority	Section / Feature	Recommendation for ISIC
High	Medicine Delivery / Buy Medicine	Full module
High	Corporate / Empanelment / Tie-ups	Full module
High	Chatbot / Virtual Assistant	Full module
Medium-High	Clinics / Network locations	Full module
Medium	Sustainability / CSR / Green initiatives	Full module
Medium	Q&A / FAQ + community Q&A	Full module
Medium	Promotional offers / Coupons / Packages	Full module
Supporting	Payment Gateway integration	Full online payments
Supporting	Video content upload & library	Full module
Supporting	Medical Guidelines / Protocols (patient-facing)	Full module

3. Detailed Development Scope - Functional Modules

Each module below corresponds to an item from the original requirements list, along with the recommended implementation phase, a technical note, and an indicative effort estimate.

3.1 Promotional Pop-up Messages

Recommended phase: Phase 1

- Configurable, non-intrusive pop-ups: timed, exit-intent, and page-specific triggers
- Support for offers, health-check discounts, new-service announcements, and events
- A/B testing, frequency capping, and click/conversion analytics
- CMS control handed to the marketing team without developer involvement

Technical note: Lightweight JS pop-up engine driven by CMS-configured rules; analytics events pushed to GA4/GTM.

Indicative effort: 4 weeks

3.2 Payment Integration

Recommended phase: Phase 1

- Gateway integration (Razorpay / PayU / CCAvenue or hospital's preferred provider)
- Coverage: appointment deposits, health-check packages, home care, medicine orders, orthodontic products, international estimates
- UPI, cards, net banking, and wallet support
- Invoice generation, refund workflow, and a reconciliation dashboard

Technical note: Server-side order/payment service with signed webhooks for payment confirmation; reconciliation ledger stored separately from the CMS.

Indicative effort: 20 weeks

3.3 Upload of Videos

Recommended phase: Phase 2

- Video CMS with categorisation: patient stories, procedures, doctor talks, "A Day at ISIC", education
- YouTube embedding plus self-hosted option
- Automatic thumbnails, captions, and mobile-optimised playback
- Playlists and in-library search

Technical note: YouTube API embed by default; self-hosted clips served via a CDN with adaptive bitrate if volume warrants it.

Indicative effort: 4 weeks

3.4 Medical Guidelines on Website

Recommended phase: Phase 2

- Patient-facing, evidence-based guidelines: condition-specific, post-operative, preventive
- Structured, readable layout with downloadable PDFs
- Version control and a clinical review/sign-off workflow
- Linked from disease/condition pages and the Health Library

Technical note: CMS content type with a review-state workflow (draft → clinical review → published) and PDF export.

Indicative effort: 4 weeks

3.5 Preventive Health Check-up Details

Recommended phase: Phase 1

- Package cards: Basic, Comprehensive, Executive, Senior, Women, Cardiac, Spine-focused, etc.
- Comparison table with inclusions, exclusions, price, and duration
- Online booking with payment
- Custom package request form

Technical note: Structured package catalogue in the CMS feeding both the comparison table and the booking flow.

Indicative effort: 8 weeks

3.6 Slots of Doctors

Recommended phase: Phase 1

- Real-time or near-real-time availability calendar per doctor
- Filters by specialty, date, time, language, and location for multi-clinic search
- Instant booking, waitlist, and reschedule
- Integration readiness with the hospital's HIS / appointment system

Technical note: This is the highest-risk item on the list because it depends on HIS API availability and data quality; recommend an early technical discovery spike before committing to a delivery date.

Indicative effort: 8 weeks (assumes HIS API is available; add 2-3 weeks if a custom bridge is needed)

3.7 Discounted Promotion Coupons

Recommended phase: Phase 2

- Coupon engine: percentage, flat, or package-specific discounts
- Unique codes, usage limits, and validity periods
- Auto-apply or manual entry at checkout
- Admin dashboard for creation and tracking
- Linked to promotional pop-ups and email/SMS campaigns

Technical note: Coupon service shared by the booking, package, and pharmacy checkouts so codes work consistently across modules.

Indicative effort: 4 weeks

3.8 Home Care Services

Recommended phase: Phase 2

- Nursing, physiotherapy, sample collection, medicine delivery, post-op care, equipment rental
- Service catalogue, booking form, pricing, and order tracking

Technical note: Booking flow module like the appointment system but with service SKUs instead of doctor slots.

Indicative effort: 8 weeks

3.9 Orthodontics Product Sell Category

Recommended phase: Phase 2

- Dedicated e-commerce category: braces, aligners, retainers, oral care products
- Product listing, detail pages, images, pricing, cart, and payment
- Doctor recommendation integration
- Inventory and order management backend

Technical note: Lightweight commerce module (product, cart, order) reusing the payment gateway from Module 2.

Indicative effort: 12 weeks

3.10 Buy Medicine

Recommended phase: Phase 2

- Online pharmacy section or integration with an existing pharmacy partner
- Search by name/composition, prescription upload, cart, payment, doorstep delivery
- Order tracking and refill reminders
- Compliance with applicable pharmacy regulations

Technical note: Scope depends heavily on whether ISIC has an existing pharmacy partner with an API versus needing a build-from-scratch catalogue; confirm this before estimating in detail.

Indicative effort: 16 weeks if integrating a partner API; 7+ weeks if building the pharmacy catalogue from scratch

3.11 Corporate Tie-ups Details

Recommended phase: Phase 2

- Directory of corporate partners, benefits, empanelled services, and enrolment process
- Search/filter by company name

- Lead form for new corporate enquiries
- Employee self-service booking, if integrated

Technical note: Standard CMS-driven directory with a CRM-connected enquiry form.

Indicative effort: 6 weeks

3.12 PSU Tie-up Details

Recommended phase: Phase 2

- Dedicated section for Public Sector Undertakings (CGHS, ECHS, and specific PSUs)
- Eligibility, process, required documents, and contact points
- Module on Fortis/Max empanelled-scheme pages

Technical note: Shares the directory pattern from Module 11.

Indicative effort: 6weekss

3.13 A Day at ISIC

Recommended phase: Phase 2

- Immersive multimedia page or video series: admission -> diagnostics -> treatment -> rehab -> discharge
- Virtual tour elements and staff introductions
- Facility highlights for trust-building with domestic and international patients

Technical note: Primarily content production (video/photo) plus a scrollytelling-style page template.

Indicative effort: 8 weeks (site build) - excludes video production time

3.14 Chatbot Integration (Agentic AI)

Recommended phase: Phase 1

- Advanced conversational interface covering the features in the dedicated Agentic AI section below

Technical note: See the Agentic AI Capabilities section for the technical approach.

Indicative effort: See Agentic AI section

3.15 All Types of Booking Facilities

Recommended phase: Phase 1

- Unified hub: doctor appointments, health checks, diagnostics/lab tests, home care, procedure enquiries, second opinion, international packages, tele-consultations
- Single dashboard for patients to manage all bookings

Technical note: A shared booking framework (one data model, one dashboard) that Modules 5, 6, 8, and 20 all plug into, rather than separate one-off booking forms.

Indicative effort: 20 weeks (framework) + time already counted in each individual booking module

3.16 Find Disease and Conditions by First Letter

Recommended phase: Phase 1

- A-Z index, similar to Apollo's "Find Diseases & Conditions By Alphabet"
- Each condition page: overview, symptoms, causes, diagnosis, treatment at ISIC, related doctors, FAQs, and a book-now CTA

Technical note: CMS content type with an auto-generated alphabetical index and cross-links to doctors and packages.

Indicative effort: 8 weeks

3.17 Blogs

Recommended phase: Phase 1

- SEO/AEO/GEO-optimised blog with categories and doctor author bios
- Related posts and social sharing
- Editorial calendar and a straightforward publishing workflow

Technical note: Standard CMS blog module with structured metadata for search and AI-answer-engine visibility.

Indicative effort: 15 blogs in a month of 2500 words (ongoing)

3.18 Q&A Section

Recommended phase: Phase 2

- Structured FAQ by specialty plus free-text search
- User-submitted questions, moderated by the clinical team
- Public answers feeding AEO/GEO visibility

Technical note: Shares infrastructure with Module 22 (Quora-style Q&A) — recommend building them together.

Indicative effort: 25 Q&A section in a month (ongoing)

3.19 Search - Disease, Treatment, Doctor

Recommended phase: Phase 1

- Unified intelligent search (semantic + keyword) across diseases, treatments, doctors, packages, and blogs
- Autocomplete, filters, and ranked results

Technical note: A dedicated search index (e.g. Algolia/Elasticsearch) fed from every content type, rather than relying on native CMS search.

Indicative effort: 16 weeks

3.20 Request a Call Back

Recommended phase: Phase 1

- Prominent forms in header, footer, specialty pages, and on exit-intent
- Separate domestic and international flows
- CRM integration with SLA tracking

Technical note: Form submissions routed directly into the hospital's CRM with SLA timers configured per flow.

Indicative effort: 12 week

3.21 Health Library

Recommended phase: Phase 2

- Central repository of articles, videos, guidelines, and infographics
- Filterable by specialty, audience (patient/caregiver/professional), and format

Technical note: Aggregation layer over Modules 3, 4, and 17 content types rather than a separate content store.

Indicative effort: 8 weeks

3.22 Quora Q&A Style

Recommended phase: Phase 2

- Curated or community-style Q&A section
- Doctors answer popular questions; content optimised for search and AI citation
- Optional moderated user-question submission

Technical note: Built together with Module 18 — see note above.

Indicative effort: Included in Module 18 estimate

3.23 Symptoms Guide

Recommended phase: Phase 2

- Interactive or structured symptom checker with prominent disclaimers
- Maps symptoms to possible conditions and recommended next steps / booking

Technical note: Rules-based checker initially; can later be handed to the Symptom & Condition Guidance Agent described below.

Indicative effort: 12 weeks

3.24 Sustainability

Recommended phase: Phase 3

- Green initiatives, energy efficiency, waste management, community programmes, ESG commitments

Technical note: Static CMS page; no special technical requirements.

Indicative effort: 4 weeks

3.25 Clinics

Recommended phase: Phase 2

- Directory of ISIC clinics and outreach centres with maps, services, and doctors
- Clear differentiation from the main hospital
- Booking entry point per clinic

Technical note: Directory pattern shared with Modules 11/12, with a map integration (Google Maps).

Indicative effort: once in a month (ongoing)

4. Agentic AI Capabilities- ISIC IPR – Overall 40 weeks

Rather than a simple scripted chatbot, ISIC's site should be powered by agentic AI systems capable of reasoning, planning, using tools, and completing multi-step tasks on the patient's behalf, within strict clinical safety guardrails and disclaimers.

4.1 Recommended Agentic Modules

4.1.1 Primary Conversational Agent (Site-wide Chatbot)

- Handles FAQs, navigation, basic triage information, package recommendations, and booking initiation
- Multi-lingual: English, Hindi, and key international languages

4.1.2 Appointment & Booking Agent

- Understands natural language requests (e.g. "I need a spine specialist next Tuesday afternoon")
- Checks real-time slots, suggests alternatives, collects details, confirms booking, sends calendar invite / SMS

4.1.3 Symptom & Condition Guidance Agent

- Collects symptoms conversationally and maps them to possible conditions using an approved medical knowledge base
- Explains next steps and offers to book the relevant specialist or health check
- Always shows a clear "not a diagnosis" disclaimer with escalation to a human

4.1.4 International Patient Concierge Agent

- Guides the journey from enquiry to estimate request, document checklist, visa support, travel planning, and airport pickup coordination
- Triggers human handoff and creates CRM tickets when needed

4.1.5 Personalised Health Journey Agent

- Remembers user context, with consent, to recommend relevant packages, doctors, blogs, and follow-up care
- Provides post-discharge support: home-care scheduling and medicine reminders

4.1.6 Content & Knowledge Agent (Internal + Public)

- Powers Health Library search
- Generates draft FAQs or blog outlines for human review
- Helps keep disease pages up to date

4.1.7 Corporate / PSU Enquiry Agent

- Qualifies corporate leads, explains empanelment benefits, and routes qualified leads to the business development team

4.2 Technical Approach

- LLM backbone with retrieval-augmented generation (RAG) over ISIC's approved content, guidelines, doctor data, and packages
- Tool-calling / function-calling architecture so the agent can query slots, create bookings, generate payment links, and update the CRM
- Session and memory management, scoped to what the patient has consented to share
- A strict safety layer: medical disclaimers, escalation rules, content filtering, and audit logs
- Analytics on conversation success rate, booking conversion, and common intents
- A continuous improvement loop with human-in-the-loop review of agent responses

5. Recommended Implementation Priority

The scope is grouped into three delivery phases, sequencing critical patient-facing and revenue-driving capabilities first.

Phase 1 - Core Launch

Doctor slots and all core booking types, health-check packages, disease A-Z, unified search, request callback, basic chatbot, payment integration, blogs, promotional pop-ups.

Indicative duration: ~36 weeks

Phase 2

Home care, buy medicine, orthodontics products, corporate & PSU pages, video library, "A Day at ISIC", medical guidelines, symptoms guide, Q&A / Quora-style Q&A, coupon engine, Health Library, clinics directory.

Indicative duration: ~28 weeks (varies significantly on pharmacy scope - see Module 10)

Phase 3

Full agentic AI suite (beyond the basic chatbot), advanced personalisation, sustainability page, coupon engine maturity, clinics directory enhancements.

Indicative duration: ~42 weeks, dependent on LLM/RAG platform selection

6. Overall Effort & Timeline Summary

The table below consolidates the indicative phase durations. These are elapsed-time estimates assuming a small dedicated team (roughly 2-3 developers, 1 designer, 1 QA, part-time PM) working in parallel across modules within a phase; they are not simple sums of the individual module estimates in Section 3, since many modules within a phase are built concurrently.

Phase	Scope summary	Indicative duration
Phase 1 - Core Launch	Booking, health checks, disease A-Z, search, callback, basic chatbot, payments, blogs, pop-ups	14-16 weeks
Phase 2	Home care, pharmacy, orthodontics store, corporate/PSU, video, guidelines, symptoms guide, Q&A	18-22 weeks
Phase 3	Full agentic AI suite, personalisation, sustainability, coupon maturity, clinics enhancements	42 weeks

Total indicative timeline across all three phases: approximately 42-52 weeks (10-12 months) of elapsed delivery time, run largely back-to-back with some overlap between phases. The two biggest swing factors are the HIS integration for real-time doctor slots (Module 6) and the scope of the pharmacy module (Module 10) - both should be validated early, as they can materially move the Phase 1 and Phase 2 timelines.

7. Assumptions & Next Steps

7.1 Key Assumptions

- Estimates assume a small dedicated delivery team working across all three phases, not a single generalist.
- The hospital's HIS exposes (or can be made to expose) an API for doctor availability and booking; if not, a bridging integration will add time to Phase 1.
- A pharmacy partner with an existing API is assumed for the fastest path on Module 10; building a pharmacy catalogue from scratch will extend Phase 2.
- Content (doctor bios, package details, video assets, medical guideline copy) is supplied or reviewed by ISIC's clinical and marketing teams on a schedule that keeps pace with development.
- Agentic AI features (Section 4) assume access to an LLM provider and are scoped separately from the core CMS/booking build.

7.2 Suggested Next Steps

- Confirm HIS and pharmacy-partner API availability with early technical discovery spikes.
- Review and reprioritise the module list in Sections 3 and 5 against ISIC's actual budget and timeline constraints.

- Produce wireframes for the Phase 1 modules to firm up the effort estimates into committed figures.
- Select the LLM/RAG platform for the agentic AI suite ahead of Phase 3 planning.